

Logistic Population Growth Model:

Let t = the time the population grows.

P or $P(t)$ = the population after time t ,

K = relative growth rate coefficient,

M = Carrying Capacity, the amount ^{that} when exceeded will result in the population to decrease

P_0 = Initial population, or the population we start with at time $t=0$, that is; $P(0) = P_0$

The initial value problem for logistic population growth is given by:

$$\frac{dp}{dt} = kP \left(1 - \frac{P}{M} \right), \quad P(0) = P_0$$

has a particular solution:

$$P(t) = \frac{M}{1 + A e^{-kt}}, \quad \text{where } A = \frac{M - P_0}{P_0}$$

More specific:

$\frac{dp}{dt}$ = Rate of Population change.

k = Maximum Population growth rate.

P = Population size

M = Population carrying capacity i.e. the maximum population size of a biological species that can be sustained by specific environment.

$$\begin{array}{c} \text{Birth} \\ \downarrow \\ B \end{array} - \begin{array}{c} \text{Death} \\ \downarrow \\ D \end{array} = \left(1 - \frac{P}{M}\right)$$

$$\left. \begin{array}{l} B = \text{born} \\ D = \text{death} \end{array} \right\}$$

$$\frac{dp}{dt} = k(B - D) p(t)$$

$$\frac{dp}{dt} = k \left(1 - \frac{p}{M}\right) p$$

Compare by $\frac{dy}{dt} = k \left(1 - \frac{y}{M}\right) y$

$$\frac{dy}{\left(1 - \frac{y}{M}\right) y} = k dt$$

$$\int \frac{dp}{p \left(1 - \frac{p}{M}\right)} = k dt$$

$$\int \frac{dp}{p \left(1 - \frac{p}{M}\right)} = kt + C$$

If we use partial fractions we can get:

$$p(t) = \frac{CM e^{kt}}{1 + C e^{kt}}, \quad \text{where } C = \frac{M - p_0}{p_0}, \quad \boxed{M > p}$$

$\downarrow$
 Growth.

(4)

EX Suppose a species of fish in a lake is modeled by a Logistic Population model with relative growth rate of $k = 0.3$ Per year and carrying capacity of $M = 10000$.

(a) Write the differential equation describing the Logistic Population model for this problem.

Note 1) $P(t) = \frac{C M e^{kt}}{1 + C e^{kt}}$

$$P(t) = \frac{\left(\frac{M-P_0}{P_0}\right) M e^{kt}}{1 + \left(\frac{M-P_0}{P_0}\right) e^{kt}} \longrightarrow P(t) = \frac{\frac{10,000-P_0}{P_0} (10,000) e^{0.3t}}{1 + \frac{10,000-P_0}{P_0} (10,000) e^{0.3t}}$$

Suppose

$P(0) = 5000$, then

$$C = \frac{M-P_0}{P_0} = \frac{10,000 - 5000}{5000} = \boxed{1}$$

$$\Rightarrow \boxed{C = 1}$$

$$(a) P(t) = \frac{C M e^{kt}}{1 + C e^{kt}} = \frac{(10,000) e^{0.3t}}{1 + 1 e^{0.3t}} = \boxed{\frac{10,000 e^{0.3t}}{1 + e^{0.3t}}}$$

Use the formula in (a) to estimate the number of fish in the lake after 5 years.

Solution $P(5) = \frac{10,000 e^{(0.3)(5)}}{1 + 1 e^{(0.3)(5)}} = \frac{10,000 e^{(1.5)}}{1 + 1 e^{(1.5)}}$

$$P(5) = \frac{44816.89}{5.482} = 8125.28 \approx 8125 \text{ (fish)}$$

(c) Estimate the time it will take for there to be 8000 fish in the lake.

Solution $P(t) = \frac{10000 e^{0.3t}}{1 + 1 e^{0.3t}}$ - Find t if $P = 8000$

So, $\frac{8000}{1} = \frac{10000 e^{0.3t}}{1 + 1 e^{0.3t}} \Rightarrow 8000 + 8000 e^{0.3t} = 10,000 e^{0.3t}$

Note $10000 - 8000 = 2000$

$$(2000) e^{0.3t} = 8000$$

$$e^{0.3t} = \frac{8000}{2000} = 4$$

$$0.3t = \ln(4)$$

$$t = \frac{\ln(4)}{0.3}$$

$$t = 4.62 \text{ years}$$

(6)

Rate of Temperature Proportional to The difference between the temperature of the body and the temperature of the surrounding.

Let $T(t)$ be the temperature at any time t to the surrounding temperature.

$$\frac{dT}{dt} \propto T(t) - T_0$$

→ proportional relation

$$\frac{dT}{dt} = k(T - T_0)$$

Temp of body
Temp of surrounding

$T(0) = A$ (The initial temperature)

$$\frac{dT}{T - T_0} = k dt \xrightarrow{\text{integrate}} \ln |T - T_0| = kt + C$$

$(T > T_0)$

$$e^{\ln |T - T_0|} = e^{kt + C} \Rightarrow$$

$$T - T_0 = C e^{kt} \quad (T > T_0) \rightarrow \text{for cooling}$$

OR

$$T(t) = T_0 + C e^{kt}$$

$$-(T - T_0) = T_0 - T$$

→ for heating

Ex #1) Suppose $T(0) = 90^\circ$, $T_0 = 20^\circ$, $T(5) = 80^\circ$ ⑦

$$T(t) = T_0 + C e^{kt}$$

$$T(0) = 20 + C e^0$$

$$90 = 20 + C \Rightarrow C = 90 - 20 = \boxed{70}$$

$$\therefore T(t) = 20 + 70 e^{kt}$$

Suppose after 5 hours the temperature was 80, then

$$T(5) = 20 + 70 e^{5k}$$

$$80 = 20 + 70 e^{5k}$$

$$70 e^{5k} = 80 - 20$$

$$e^{5k} = \frac{60}{70}$$

$$5k = \ln\left(\frac{6}{7}\right) \Rightarrow k = \frac{\ln\left(\frac{6}{7}\right)}{5} = \boxed{-0.03}$$

$$\therefore T(t) = 20 + 70 e^{-0.03t}$$

Find t when $T = 60$?

$$60 = T(t) = 20 + 70 e^{-0.03t}$$

$$70 e^{-0.03t} = 40 \Rightarrow e^{-0.03t} = \frac{40}{70}$$

$$t = \frac{\ln\left(\frac{4}{7}\right)}{-0.03} = 18.6 \text{ ~~minutes~~ minutes.}$$

Ex #2 Suppose $T(0) = 200$, $T_0 = 80$,
and $T(30) = 120$? Find $T(60)$?

$$T(t) = T_0 + Ce^{kt}$$

$$T(0) = 80 + Ce^0$$

$$200 = 80 + C \Rightarrow \boxed{C = 120}$$

$$\therefore T(t) = 80 + 120e^{kt}$$

$$T(30) = 80 + 120e^{30k}$$

$$120 = 80 + 120e^{30k}$$

$$120e^{30k} = 40$$

$$e^{30k} = \frac{40}{120} = \frac{1}{3}$$

$$30k = \ln\left(\frac{1}{3}\right)$$

$$k = \frac{\ln\left(\frac{1}{3}\right)}{30} = -0.04$$

$$\therefore T(t) = 80 + 120e^{-0.04t}$$

we need to find $T(60)$?

$$T(60) = 80 + 120e^{(-0.04)(60)}$$

$$= 80 + 120e^{-2.4}$$

$$T(60) = 90.9^\circ$$